THE CROSS TERM OF THE REDUCED CUP SQUARE, AND A DELOOPING OF $\mathrm{Fib}\bar{\kappa}$

Purpose and status. First task of §2 of `doc/pi3-s2-postnikov-handoff-9.txt` (towards $\pi_4(S^3) = \mathbb{Z}/2$): an informal proof that the cross term of $\bar{\kappa} = \kappa; K(\rho, 4)$ vanishes, and that $R = \mathrm{Fib}\bar{\kappa}$ consequently deloops, with a homogeneous structure and a torsor as corollaries. Written before any code; the numbered statements are meant to be declarations (§8). Foundations as in the development (univalence, images; function extensionality derived). No new axiom; *the S^3 axiom is not used* (it is needed only for step 3 of the handoff); no theory of quadratic maps, no pushout, no smash product. Refereed twice (28 Oct; second review of the revision: no further obstruction): main route sound; repairs incorporated (Lemma 3 for PMO_1 with arbitrary A, in place of `PICircle`; τ as an explicit pointed equivalence through `PtMapLoops`, Lemma 11; the converse in §7 needs a normalisation of the cell and is withdrawn as open; Lemma 8 with the locks of e_1 and of the circle generator made explicit; citations corrected).

The idea. Write $K_m = K(\mathbb{Z}, m)$, $K'_m = K(\mathbb{Z}/2, m)$, $X = K_3$, $W = K'_5$, and let $f = e_2; \bar{\kappa}; e_4^{-1} : \Omega X \to_* \Omega W$ be $\bar{\kappa}$ seen on loop spaces. By DW1's Lemma 9 (`DeloopMaps`), f is the looping of a map $\hat{F} : X \to_* W$ as soon as the cell at the base point is inhabited, and at $c = 2$, $t = 1$ the cells are propositions. So *no pointing in p is needed*: it suffices to have, for every $p : \Omega X$, a path $f = D_p$ in PMO, where $D_p(q) = (fp)^{-1} \cdot f(p \cdot q)$. These paths come from the *cross term*, the bipointed map $\tilde{\delta}(x)(y)$ "$= \bar{\kappa}(x + y) - \bar{\kappa}(x) - \bar{\kappa}(y)$", which we show is 0 by Theorem A of `doc/h4.pdf`: its diagonal is "$\bar{\kappa}(2x) - 2\bar{\kappa}(x)$", and both terms vanish *by construction* (not as classes) because $2 = 0$ in $\mathbb{Z}/2$. Then $BR = \mathrm{Fib}\hat{F}$ is a pointed 2-connected 4-type with $\Omega BR \simeq_* R$.

Where $2 = 0$ enters, twice. (i) Doubling on $\Omega K'_{m+1}$ is $e_m; K(2, m); e_m^{-1}$ (Lemma 6), and $K(2, m) = K(0, m) = 0$ over $\mathbb{Z}/2$. (ii) Precomposing c_2 in the first variable with $K(2, 2)$ multiplies its invariant $\mathcal{E}$ by 2 (Lemma 8); after $K(\rho, 4)_*$ this is again doubling in $\Omega^2 K'_4$, which is 0 (Theorem 10). Over $\mathbb{Z}$ neither vanishes (the general formula, Proposition 21, gives the cross term $c_2; K(2g, 4)_*$).

1. Conventions

As in `doc/h4.pdf`: `isTrunc n` is "$(n-2)$-truncated", `isConn n` is "$(n-2)$-connected"; $K(A, m) =$ `KK[A].K m` with $e_m : \Omega K(A, m+1) \simeq_* K(A, m)$ (`KK.e`, e_0 the band); $F; G$ is `ptcomp F G`; $p \cdot q$ is `trans p q` (so $p \cdot \mathsf{refl} \equiv p$, $\mathsf{refl} \cdot \mathsf{refl} \equiv \mathsf{refl}$); for pointed X, Y, $\mathrm{Map}_*(X, Y)$ is pointed by $0 = (\lambda_.y_0, \mathsf{refl})$; $M_Y = \mathrm{Map}_*(K_2, Y)$; loop, D, Kf, E, Unloop, Rel, Levels are those of `DeloopMaps.Maps`; $\mathsf{PMO}_m = \mathrm{Map}_*(\Omega K(A, m+1), \Omega K(A, m+1))$; $\mathrm{conj}_m F = e_m; F; e_m^{-1}$ (`KMap.conj`), so $\mathsf{loop} K(h, m+1) = \mathrm{conj}_m K(h, m)$ (`loopK`). From `doc/h4.pdf`: $c_2 : K_2 \to_* M_{K_4}$, $\kappa = \Delta c_2$ (by definition), $\mathcal{E}$ and $\Theta_B = \mathcal{E}; j_B$ (`H4Bilinear.EDef`, `BilAt.Theta`), Theorem A, Theorem B, Lemma j, the pointing irrelevance lemma `pi-set`.

2. The group $\mathbb{Z}/2$ and the data

Definition 1. $\mathbb{Z}/2$ is the abelian group (`Torsor.Ab`) with carrier `Bool`, $o = \mathsf{false}$, $a + b = \mathsf{xor}(a, b)$, $-a = a$; it is a set (decidable equality, `Hedberg`); the laws hold by cases. For an abelian A, $2_A : \mathrm{Hom}(A, A)$ is $a \mapsto a + a$ (associativity and commutativity), $0_{A,B} : \mathrm{Hom}(A, B)$ is constant o. The

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reduction $\rho : \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}/2)$ is the parity, with $\rho(\mathsf{suc}\, x) = \rho(\mathsf{pred}\, x) = \mathsf{not}(\rho\, x)$, so $\rho(x+y) = \rho x + \rho y$ by induction on y, as in `Integers`.

Lemma 2. $2_{\mathbb{Z}/2} = 0_{\mathbb{Z}/2,\mathbb{Z}/2}$ *in* $\mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2)$.

Proof. $\mathsf{xor}(a, a) = \mathsf{false}$ by cases; the homomorphism parts agree because the carrier is a set. □

Let $\bar{\kappa} = \kappa; K(\rho, 4) : K_2 \to_* K_4'$ and $R = \mathrm{Fib}\bar{\kappa}$. By `PostnikovRealization.Triple` at $(\mathbb{Z}, \mathbb{Z}/2, \bar{\kappa})$, R is `isConn 3`, `isTrunc 5`, and $\pi_3 R \cong \mathbb{Z}/2$ as groups.

3. Doubling on Eilenberg–MacLane loop spaces

For a pointed type (Y, y_0) let $\mathsf{sq} : \Omega Y \to_* \Omega Y$, $\mathsf{sq}(p) = p \cdot p$, pointed by refl ($\mathsf{refl} \cdot \mathsf{refl} \equiv \mathsf{refl}$).

Lemma 3 (pointing irrelevance, 1-types)**.** *Let* (P, p_0) *be* ***`isConn 2`*** *and* (Q, q_0) ***`isTrunc 3`*** *with* $\Omega(Q, q_0)$ *commutative. Then* $(g, \zeta) = (g, \zeta')$ *for all* $g : P \to Q$ *and* $\zeta, \zeta' : g\, p_0 = q_0$.

Proof. By `unique-loop1` ($c = 0$; $3 = 1 + (1 + 1)$) it suffices that $\mathsf{loop}(g, \zeta) = \mathsf{loop}(g, \zeta')$ in $\mathrm{Map}_*(\Omega P, \Omega Q)$. By `loop-ap` the underlying maps are $q \mapsto \zeta^{-1} \cdot \mathsf{ap}_g q \cdot \zeta$ and the same with ζ'; they differ by conjugation by the loop $\zeta^{-1} \cdot \zeta'$ of Q (`LoopConj.conj-trans`), which is trivial since ΩQ is commutative. The pointings lie in identity types of the set ΩQ, which are propositions. □

Lemma 4 (pointing irrelevance on PMO_m)**.** *For every* m, PMO_m *is a set, and two elements of* PMO_m *with pointwise equal underlying maps are equal.*

Proof. Set: $m = 0$, the target is the set $\Omega K(A, 1)$; $m \geq 1$, `cor5-map` (source `isConn (m+1)`, target `isTrunc (m+2)`). Pointings: two pointings of one map differ by a loop of $\Omega K(A, m{+}1)$ at refl, i.e. an element of $\Omega^2 K(A, m{+}1)$. $m = 0$: this type is a proposition. $m = 1$: Lemma 3 with $P = Q = \Omega K(A, 2)$ (2-loops commute, `EH.eh`); `PICircle` is not used, it is the circle property of $K(\mathbb{Z}, 1)$ only. $m \geq 2$: $\Omega^2 K(A, m{+}1)$ is connected ($\simeq K(A, m{-}1)$) and `pi-set` applies. □

Lemma 5 (K of zero)**.** $K(0_{A,B}, m) = 0$ *for all* m.

Proof. $m = 0$: both underlying maps are constant o; pointings in a set. $m{+}1$: by `KFun.Unloop.unloop` it suffices that the loopings agree: $\mathsf{loop}K(0, m{+}1) = \mathrm{conj}_m K(0, m) = \mathrm{conj}_m 0 = 0 = \mathsf{loop}0$ (`loopK`, induction, composition with a constant map, `loop-const`). □

Lemma 6 (doubling)**.** *For every abelian* A *and* m, $\mathsf{sq} = \mathrm{conj}_m K(2_A, m)$ *in* PMO_m, *i.e.* $\mathsf{sq} = \mathsf{loop}K(2_A, m{+}1)$.

Proof. By Lemma 4 it suffices to compare underlying maps, by induction on m. $m = 0$: $\mathrm{conj}_0 K(2, 0) = \mathsf{unband} \circ 2 \circ \mathsf{band}$ (`c0-conj`), and $\mathsf{band}(p \cdot p) = \mathsf{band}\, p + \mathsf{band}\, p$ (`band-hom`) and $\mathsf{band}(\mathsf{unband}\, a) = a$ (`band-unband`); conclude with `band-inj`. $m + 1$: let $G = e_{m+1}^{-1}; \mathsf{sq}; e_{m+1} : K(A, m{+}1) \to_* K(A, m{+}1)$. It suffices that $G = K(2, m{+}1)$ (then $\mathsf{sq} = e_{m+1}; G; e_{m+1}^{-1}$ by `pt-inv-l/r`), hence, by `KFun.Unloop.unloop`, that $\mathsf{loop}G = \mathsf{loop}K(2, m{+}1) = \mathrm{conj}_m K(2, m)$, an equality in PMO_m, so pointwise (Lemma 4). By `loop-comp`, $\mathsf{loop}G = \mathsf{loop}e_{m+1}^{-1}; \mathsf{loopsq}; \mathsf{loop}e_{m+1}$. (a) loopsq is sq on $\Omega^2 K(A, m{+}2)$ pointwise: $(\mathsf{loopsq})(\alpha) = \mathsf{refl}^{-1} \cdot \mathsf{ap}_{\mathsf{sq}}\alpha$ and $\mathsf{ap}_{q \mapsto q \cdot q}\alpha = \alpha \cdot \alpha$ for $\alpha : \mathsf{refl} = \mathsf{refl}$ (a new lemma `ap-diag`: ap of $q \mapsto q \cdot q$ along $a : p = p'$ is $\mathsf{ap}_{- \cdot p} a \cdot \mathsf{ap}_{p' \cdot -} a$, by J; then `EH.whisk-r`, `EH.whisk-l`; `Groupoid.ap-trans` is a different statement). (b) With $\gamma = (\mathsf{loop}e_{m+1}^{-1})\beta$: $(\mathsf{loop}e_{m+1})(\gamma \cdot \gamma) = (\mathsf{loop}e_{m+1})\gamma \cdot (\mathsf{loop}e_{m+1})\gamma$ (`loop-mult`) and $(\mathsf{loop}e_{m+1})\gamma = \beta$ (`loop-comp`, `pt-inv-r`, `loop-id`). So $(\mathsf{loop}G)\beta = \beta \cdot \beta = \mathsf{sq}\, \beta$, which is $(\mathrm{conj}_m K(2, m))\beta$ by induction. □

Corollary 7. *Over* $\mathbb{Z}/2$: $\mathsf{sq} = 0$ *in* PMO_m *for every* m; *so there are homotopies* $s_m : \prod_{p : \Omega K'_{m+1}} p \cdot p = \mathsf{refl}$. *Also, for* $w : \Omega^2 K_4'$, $w \cdot w = \mathsf{refl}$ *(a homotopy* s'*).*

Proof. Lemma 6, Lemma 2 (apH of $K(-,m)$), Lemma 5, and $\mathrm{conj}_m 0 = 0$. For s': $\mathsf{loop}e_3 : \Omega^2 K_4' \to_* \Omega K_3'$ is an equivalence with $(\mathsf{loop}e_3)(w \cdot w) = ((\mathsf{loop}e_3)w)^2$ (`loop-mult`); apply s_2; $(\mathsf{loop}e_3)\mathsf{refl} = \mathsf{refl}$ is the pointing of $\mathsf{loop}e_3$, and injectivity of $\mathsf{loop}e_3$ comes from its being an equivalence (`LoopIso.loopPE`). □

4. Precomposition with $K(2,2)$

Lemma 8. *For pointed Y and $c : K_2 \to_* M_Y$, the underlying map of $\mathcal{E}(K(2_\mathbb{Z},2);c)$ is pointwise $x \mapsto \mathcal{E}(c)(x) \cdot \mathcal{E}(c)(x)$.*

Proof. $\mathcal{E}(c) = \mathsf{ev}(\mathsf{evPt}(e_1^{-1}; \mathsf{loop}c; \mathsf{ev}_Y))$, with e_1 = `CupSquare.e1`, which is `KK[Zab].e 1` *by definition* but locked: a new lemma `e1-def` : e_1 = `KZ.e (suc zero)` by refl is added to `CupSquare` just before `opaque e1` (rule 3; no statement changes). By `loop-comp` and `KMap.loopK 1` (raw `KK.e`, not the locked `Lev.eA`), $\mathsf{loop}(K(2,2);c) = e_1; K(2,1); e_1^{-1}; \mathsf{loop}c$; with $L = e_1^{-1}; \mathsf{loop}c; \mathsf{ev}_Y$ the argument of evPt is $K(2,1); L$ (`e1-def`, `ptcomp-assoc`, `pt-inv-r`, `ptcomp-idl`). Let g_N be the generator of the instance `Circle[N]`, $N = \mathrm{Map}_*(K_2, \Omega Y)$, used by evPt in $\mathcal{E}$ (each instance locks its own g; compare `ENat.g-eq`). Then $\mathsf{evPt}(K(2,1);L) \equiv (\mathsf{loop}(K(2,1);L))(g_N) = (\mathsf{loop}L)((\mathsf{loop}K(2,1))\, g_N)$ (`evPt-loop`, `loop-comp`), and $(\mathsf{loop}K(2,1))\, g_N = g_N \cdot g_N$ by `loopK 0` and Lemma 6 at $m = 0$, $A = \mathbb{Z}$, applied at g_N (not through evPt of a different instance; formally `k21-sq`, with $K(2,1)$ written once at the types of the circle, `k21`, where $K(\mathbb{Z},1)$ of the circle is identified with `KK.K 1`). By `loop-mult` it is $\mathsf{evPt}\, L \cdot \mathsf{evPt}\, L$, and ev takes a product of loops of N to the pointwise product (`ev-ap`, `ap-trans`). □

Lemma 9. *For $G : Y \to_* Y'$, $(\mathsf{loop}^2 G)(\alpha \cdot \beta) = (\mathsf{loop}^2 G)\alpha \cdot (\mathsf{loop}^2 G)\beta$ (`loop-mult` for $\mathsf{loop}G$).*

Theorem 10 (the reduced cup square kills doubling)**.** *$c'' = K(2,2); c_2; K(\rho,4)_*$ is 0 in $\mathrm{Map}_*(K_2, M_{K_4'})$. Hence there is a homotopy $N : \prod_{x:K_2} \bar\kappa(K(2,2)\, x) = *$.*

Proof. $\Theta_{\mathbb{Z}/2}$ is injective (`BilAt.theta-inj`), and pointed maps $K_2 \to_* K_2'$ with equal underlying maps are equal (`pi-set`). Pointwise, $\mathcal{E}(c'') = \mathcal{E}(K(2,2);c_2); \mathsf{loop}^2 K(\rho,4)$ (`ENat.e-nat`, the ptcomp of the two outer maps is the one of the definition), which is $x \mapsto (\mathcal{E}(c_2)x \cdot \mathcal{E}(c_2)x)$ mapped by $\mathsf{loop}^2 K(\rho,4)$ (Lemma 8, $\mathcal{E}(c_2) = \iota$ by `cup-norm`), i.e. $x \mapsto w_x \cdot w_x$ with $w_x = (\mathsf{loop}^2 K(\rho,4))(\iota x)$ (Lemma 9); this is refl by s' (Corollary 7). Also $\mathcal{E}(0)$ is pointwise refl: `loop-const` and `comp-const-l/r` give $\mathsf{phi}\, 0 = 0$, then `evPt-loop` with `loop-const`, and $\mathsf{ev}\,\mathsf{refl} \equiv r$. $\Theta(c'')$ and $\Theta(0)$ are then pointwise the point through the pointing of j_B, and `pi-set` (`BilAt.sK2`, `cK2`) makes them equal, as in `key`. The same $K(\rho,4)$, the locked `Lev[Zab,Z2ab].K rho four` (as in `BilAt.c2g`), is used both in $\bar\kappa$ and in $K(\rho,4)_*$, so that the unfolding for N below is definitional (with the raw `KMap.K four` the checker unfolded its construction, `four` being a numeral). So $\Theta(c'') = \Theta(0)$ and $c'' = 0$. For N: by definition $c''(x) = c_2(K(2,2)x); K(\rho,4)$, so $c''(x)_1(K(2,2)x) \equiv K(\rho,4)(\kappa(K(2,2)x)) \equiv \bar\kappa(K(2,2)x)$; evaluate $c'' = 0$ at x and then at $K(2,2)x$. □

5. The cross term, and why it vanishes

From now on $X = K_3$, $W = K_5'$ (pointed by x_0, w_0), with $\mathsf{isConn}\, 4\, X$ and $\mathsf{isTrunc}\, 7\, W$. Let $f = e_2; \bar\kappa; e_4^{-1} : \Omega X \to_* \Omega W$ (`KMap.conj` shape), an element of PMO of `Maps[X,x0,W,w0]`, and $D_p = D\, x_0\, w_0\, f_1\, p : \mathsf{PMO}$ for $p : \Omega X$, so $D_p(q) = (fp)^{-1} \cdot f(p \cdot q)$ by definition.

Lemma 11 (translation in PMO)**.** *Let L = `PtMapLoops[`$\Omega X, \mathsf{refl}, W, w_0$`]`, so $\mathsf{ev} : \Omega\mathrm{Map}_*(\Omega X, W) \simeq_* \mathsf{PMO}$ with $\mathsf{ev}\,\mathsf{refl} \equiv 0$ (`ev-pt`), inverse unev (`ev-unev`). Put $\ell = \mathsf{unev}\, f$ and*

$$\tau(h) = \mathsf{ev}(\mathsf{unev}\, h \cdot \ell^{-1}), \qquad \tau_0 = \mathsf{ap}_{\mathsf{ev}}(\textit{trans-sym-r}\ \ell) : \tau f = 0.$$

Then $(\tau, \tau_0) : (\mathsf{PMO}, f) \simeq_* (\mathsf{PMO}, 0)$ (a composite of equivalences: unev, right translation by ℓ^{-1} with inverse right translation by ℓ, ev), and pointwise $\tau(h)_1(q) = h_1(q) \cdot (f_1 q)^{-1}$ (`ev-ap`: ev of

a path is its pointwise ap*;* ap *of a composite and of an inverse;* ev $\circ$ unev $=$ id*). The pointing coherence is part of the construction, not discarded:* PMO *is a 2-type. Only the pointwise formula is used later (Lemma 14).*

Definition 12 (the cross term)**.** $\tilde{\delta} = e_2^{-1}; \hat{D}; \tau; \mathrm{Cj} : K_2 \to_* M_{K_4'}$, the composite of

- $\hat{D} = (p \mapsto D_p,\ (Ef)^{-1}) : (\Omega X, \mathsf{refl}) \to_* (\mathsf{PMO}, f)$;
- $\tau : (\mathsf{PMO}, f) \simeq_* (\mathsf{PMO}, 0)$, Lemma 11;
- $\mathrm{Cj} : (\mathsf{PMO}, 0) \simeq_* (M_{K_4'}, 0)$, $h \mapsto e_2^{-1}; h; e_4$ (inverse $e_2; -; e_4^{-1}$; `comp-const-l/r`).

So, by Lemma 11 for the middle factor, $\tilde{\delta}(x)(y) = e_4\big((f\bar{x})^{-1} \cdot f(\bar{x} \cdot \bar{y}) \cdot (f\bar{y})^{-1}\big)$ with $\bar{x} = e_2^{-1}x$: "$\bar{\kappa}(x+y) - \bar{\kappa}x - \bar{\kappa}y$".

Lemma 13. *If $\tilde{\delta} = 0$ then $f = D_p$ for every $p : \Omega X$.*

Proof. At $x = e_2 p$, $\tilde{\delta}(x) = 0 = \mathrm{Cj}(\tau f)$ (pointings of τ, Cj); τ is injective as an equivalence of PMO (Lemma 11; PMO is not a set, so `pi-set` cannot be used), and Cj by `pre-inj`, `post-inj`; so $D_{\bar{x}} = f$, and $\bar{x} = e_2^{-1}(e_2 p) = p$ (`sec-ret`). □

Lemma 14 (the diagonal of the cross term)**.** *$\Delta\tilde{\delta}$ is pointwise equal to the constant map; hence $[\Delta\tilde{\delta},]_* = [0]$ in $[K_2, K_4']_*$.*

Proof. By the pointwise formula of Lemma 11 (not by unfolding: τ is built through ev/unev), $\Delta\tilde{\delta}(x) = e_4\big((fp)^{-1} \cdot f(p \cdot p) \cdot (fp)^{-1}\big)$, $p = \bar{x}$; the rest of $\tilde{\delta}$ (e_2^{-1}, $\hat{D}$, Cj) unfolds by definition. By Lemma 6 ($A = \mathbb{Z}$, $m = 2$), $p \cdot p = e_2^{-1}(K(2,2)(e_2 p))$, so $f(p \cdot p) = e_4^{-1}(\bar{\kappa}(K(2,2)(e_2 p)))$ ($e_2(e_2^{-1} z) = z$, the `sec-eq` of e_2), $= e_4^{-1}(*) = \mathsf{refl}$ (N, Theorem 10; pointing of e_4^{-1}). Then $(fp)^{-1} \cdot \mathsf{refl} \cdot (fp)^{-1} \equiv \mathsf{sq}((fp)^{-1}) = \mathsf{refl}$ (s_4, Corollary 7), and $e_4(\mathsf{refl}) = *$. For the class: a pointed map (k, ζ) with $H : k \sim \mathrm{const}$ equals $(\mathrm{const}, H(*)^{-1} \cdot \zeta)$ (`ptmap-path`), and $H(*)^{-1} \cdot \zeta$ is a loop of K_4', merely refl (K_4' is `isConn 5`); the goal is a proposition. □

Theorem 15 (the cross term vanishes)**.** *$\tilde{\delta} = 0$; so $f = D_p$ for every p.*

Proof. By Theorem A for $B = \mathbb{Z}/2$, $c \mapsto [\Delta c]$ is injective on $\mathrm{Map}_*(K_2, M_{K_4'})$ (`DiagAt.diag-inj` and `ClassesAt.cls-inj`; $\mathsf{cls}(\Delta' c) = [\Delta c]$ by definition). $\Delta 0 \equiv 0$, and $[\Delta\tilde{\delta}] = [0]$ by Lemma 14. Then Lemma 13. □

6. The delooping

Theorem 16. Unloop$f$ *is contractible.*

Proof. As `Over.Mult.cor12`, with the cell given directly. Take Levels[c=2, t=1] (isConn 4 X; isTrunc 7 W and $7 = 3 + (3 + 1)$): each $(b : W) \times C\, a\, b$ is a proposition (`cell`) and Unloop$f$ is a proposition (`cor10`). At $a = x_0$, $(w_0, f_1, \lambda p.$ Theorem 15) inhabits it, so every cell is inhabited (`conn-elim 2`; `Mult` uses `conn-elim zero`, $c = 0$), giving rel : Rel, and `lemma9.r` rel : Unloop$f$. □

Definition 17. $(\hat{F}, \gamma)$ is the centre: $\hat{F} : K_3 \to_* K_5'$, $\gamma : f = \mathsf{loop}\hat{F}$. $BR = \mathrm{Fib}\hat{F}$, pointed by $(x_0, \hat{F}_0)$.

Theorem 18 (R deloops)**.** *(i) $\Omega BR \simeq_* R$. (ii) BR is `isTrunc 6` and `isConn 4` (a pointed 2-connected 4-type).*

Proof. (i) $\Omega\mathrm{Fib}\hat{F} \simeq_* \mathrm{Fib}(\mathsf{loop}\hat{F})$ (`Fibres.omega-fib`), $= \mathrm{Fib}f$ along γ^{-1} (`fib-path`), $\simeq_* \mathrm{Fib}\bar{\kappa} = R$ along e_2, e_4 (`fib-equiv`; $f = e_2; \bar{\kappa}; e_4^{-1}$). (ii) Truncation: X is `isTrunc 5`, hence `isTrunc 6` (`trunc-suc`), and the identity types of W are `isTrunc 6`. Connectedness: BR is 0-connected (`FibConn`, with the surjectivity on loops as in `surj-triv`), and $\Omega BR \simeq R$ is `isConn 3`; `conn-from-point 2` gives `isConn 4`. □

Corollary 19 (homogeneous structure, torsor)**.** *R has a homogeneous structure $\mu : (r : R) \to R \simeq_* (R, r)$, hence a torsor* $\mathsf{homog}\,\mu : \mathsf{Tor}_R$ *(`Deloop.Torsors[R,r0]`), so* PT *is contractible (`cor8`, $c = 1$: R is `isConn 3`, `isTrunc 5`=`isTrunc (2+(2+1))`), and* Tor_R *is a connected delooping of R (`theorem2`, `tor-conn`, `tor-trunc`).*

Proof. On ΩBR, $y \mapsto (q \mapsto y \cdot q)$ is a pointed equivalence $(\Omega BR, \mathsf{refl}) \simeq_* (\Omega BR, y)$ ($y \cdot \mathsf{refl} \equiv y$; inverse $y^{-1} \cdot -$). Transport along (i): for $\phi : R \simeq_* \Omega BR$, $\mu(r) = \phi; (\phi r \cdot -); \phi^{-1}$, pointed by `ret-eq`. □

Remark 20 (direct homogeneity). Corollary 19 can also be read off Theorem 15 without $\hat F$: on Fib$f$, $T_{(p,\alpha)}(q, \beta) = (p \cdot q,\ \mu_{p,q} \cdot \mathsf{ap}_2(\alpha, \beta))$, with $\mu_{p,q} : f(p \cdot q) = fp \cdot fq$ from $f = D_p$, and $T(\mathsf{refl}, f_0) = (p, \alpha)$ from the pointing part of $f = D_p$. Only the pointing in q is used; the delooping above uses even less (no pointing in p). We use Theorem 18, which also gives BR as a fibre $K(\mathbb{Z}, 3) \to K(\mathbb{Z}/2, 5)$, a presentation suited to step 3.

7. The general formula (not used)

Proposition 21 (cross-term formula)**.** *For $g : \mathrm{Hom}(\mathbb{Z}, B)$ and $F = \kappa; K(g, 4)$, the cross term $\tilde\delta_F$ (Definition 12 with B for $\mathbb{Z}/2$) is $c_2; K(2g, 4)_*$. Hence $e_2; F; e_4^{-1}$ deloops if $2g = 0$. (The converse, which would show that κ itself does not deloop, is* not *claimed here; see the end of the sketch.)*

Sketch. As Lemma 14, $[\Delta\tilde\delta_F] = [F \circ K(2, 2)] - 2[F]$ in the group $[K_2, K(B, 4)]_*$ (pointwise addition through e_4; this group structure is not in the library and must be built). Further lemmas: $K(g + h, m)$ is the pointwise sum (Lemma 6 with two homomorphisms, so Φ_B is additive); $\mathcal{E}(c; (G^*)_*) = \mathcal{E}(c) \circ G$ (precomposition in the second variable), which with Lemmas 8 and j gives $\kappa \circ K(2, 2) = \kappa; K(4, 4)$; and `KFun`. Then $[\Delta\tilde\delta_F] = \Phi_B(4g) - \Phi_B(2g) = \Phi_B(2g) = [\Delta(c_2; K(2g, 4)_*)]$, and Theorem A. *Converse (open).* An inhabitant of $\mathsf{Unloop}f$ gives, through `lemma9`, a cell (b, h, k) at x_0 with b and h not known to be w_0 and f_1; moving it to (w_0, f_1) goes through the singleton contractions of `Levels` (`s1`–`s3`) and the coherence `coh`, and the pointing parts of the resulting paths $f = D_p$ live in identity types of $\Omega^2 W$, not in a set (unlike `Mult`, where $c = 0$). Alternatively, $f = \mathsf{loop}\hat F$ is multiplicative by `loop-mult`, but the pointing coherence at $q = \mathsf{refl}$ must then be proved by J. Neither is needed for $\pi_4(S^3)$ at this stage; if step 3 needs necessity, it will be stated and proved separately. (Once normalised, the conclusion $\tilde\delta_F = 0$ from its pointwise vanishing uses `pi-set`: $\mathrm{Map}_*(K_2, M_{K(B,4)})$ is a set, and $\Omega M_{K(B,4)} \simeq \mathrm{Map}_*(K_2, K(B, 3))$ is connected because its set of components $[K_2, K(B, 3)]_*$ is trivial by (H3).) □

8. Correspondence with the formal proof

- `Z2` (new, built first): `Z2ab`, `xor` laws, `z2-set`, `rho`, `rho-hom`, `twoH`, `zeroH`, `two-zero` (checked; `z2.opens`) (Definition 1, Lemma 2).
- `Doubling[A]`: `pi1` (Lemma 3), `pmo-set`, `pmo-pi` (Lemma 4), `K0` (Lemma 5), `sq`, `ap-sq`, `dbl` (Lemma 6); at $\mathbb{Z}/2$: `sq0`, `s`, `s'` (Corollary 7).
- `CrossTerm` (checked; `cross.opens`): `LMult.lmult` (`loop-mult` with its statement written), `S'.s'` (Corollary 7, end); `k21`, `loop-k22`, `k21-sq`, `ETwo.phi-two`, `ETwo.e-two` (Lemma 8), `ETwo.e-zero` ($\mathcal{E}(0)$ pointwise refl), `l2-mult` (Lemma 9), `kbar`, `Kill.kill`, `Kill.N`, `Kill.N-lev` (Theorem 10); in `Cross`: `f`, `Dp`, `Dhat`, `tau-fun`, `tau-def`, `unev-ev`, `tau`, `tau-pt`, `tau-formula` (Lemma 11; e_2, e_4 are the locked `Lev` ones), `Cj`, `delta`, `delta-def` (Definition 12); `cell-of` (Lemma 13), `f-sq`, `diag-null`, `diag-cls` (Lemma 14: an equality of *classes* of diagonals), `cross`, `cross-cell` (Theorem 15: Theorem A's injectivity upgrades it to an actual equality $\tilde\delta = 0$ of bipointed maps, which `cell-of` turns into the paths $f = D_p$; no representative is chosen).

- `DeloopR` (checked; `deloop.opens`): `cell0`, `rel`, `unloop-contr` (Theorem 16), `Fhat`, `gamma`, `BR` (Definition 17), `sq-f`, `omega-BR`, `BR-trunc`, `BR-conn` (Theorem 18), `phi`, `transl`, `mu`, `tor`, `pt-contr`, `tor-loops`, `tor-conn`, `tor-trunc` (Corollary 19).

Each declaration has an entry in `cross.opens` or `deloop.opens`.